Skills Summary

Choosing the Form That Shows the Structure

Use this reference while completing your worksheet or when studying.

One expression, two forms. $(x-4)(x-5)$ and $x^2-9x+20$ are the same expression written two ways. Neither is "simpler" — they answer different questions.

Expanded (a sum of terms) shows: the degree, the leading coefficient, and the constant term (the value at $x=0$).

Factored (a product) shows: the zeros, and any factor shared by every term.

So read the *question* first, then choose the form that puts the answer in plain sight.

1. Expand a product to reveal its terms

Every term of the first factor multiplies every term of the second, then like terms are collected. Useful patterns: $(A+B)^2=A^2+2AB+B^2$ and $(A-B)(A+B)=A^2-B^2$.

Example: Write $(x+4)(x+1)$ in expanded form.
Step 1. The question asks for the terms, so the product has to be distributed.
Step 2. $x\cdot x=x^2$; the cross terms are $x\cdot 1+4\cdot x=5x$; the last product is $4\cdot 1=4$.
Collecting gives $\boldsymbol{x^2+5x+4}$, so the constant term is 4.

Try it: Write $(x+6)(x+2)$ in expanded form.

check: $\boldsymbol{x^2+8x+12}$

2. Factor a sum to reveal its structure

Pull out the greatest common factor first, then look for a pattern. For x^2+bx+c, find two numbers with product c and sum b. No middle term and two perfect squares means $A^2-B^2=(A-B)(A+B)$.

Example: Factor x^2-5x+6 completely.
Step 1. The zeros are wanted, so the expression must be turned into a product.
Step 2. Two numbers with product 6 and sum -5 are -2 and -3, and there is no common factor to remove first.
So $x^2-5x+6=\boldsymbol{(x-2)(x-3)}$, which shows zeros at 2 and 3.

Try it: Factor x^2-25 completely.

check: $\boldsymbol{(x-5)(x+5)}$

3. Choose the form that answers the question

Asked for a zero, or "when is this 0?" — use the **factored** form and set one factor to zero. Asked for a value, a constant term or a cancellation — use the **expanded** form. A hard product of numbers is often an A^2-B^2 in disguise.

Example: Find $(n-2)(n+2)$ when $n = 50$.
Step 1. Multiplying 48 by 52 by hand is slow, and the product matches $A^2 - B^2$, so expanding is the cheaper route here.
Step 2. $(n-2)(n+2) = n^2 - 4$, and at $n = 50$ that is $2500 - 4$.
The value is **2496**.

Try it: Find $(n-5)(n+5)$ when $n = 40$.

check: **1575**

(!) Watch out: $(2x-3)^2$ is not $4x^2 + 9$ — squaring a binomial always leaves a middle term. And "factor completely" means the greatest common factor comes out *before* you hunt for a pattern: $2x^3 - 18x$ is $2x(x-3)(x+3)$, not $(2x^2 - 18)(x)$ left half-done.